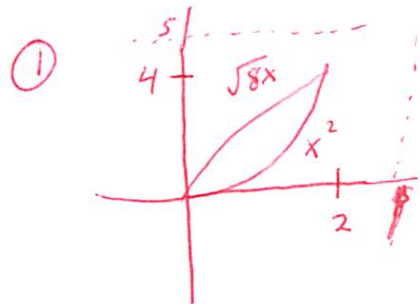




④ $\int_0^2 \sqrt{8x - x^2} dx$ or $\int_0^4 \sqrt{y} - \frac{1}{8} y^2 dy$

③ $\int_0^2 2\pi x (\sqrt{8x - x^2}) dx$ shell $\int_0^4 \pi (\sqrt{y})^2 - \pi (\frac{1}{8} y^2)^2 dy$ washer

⑤ $\int_0^2 \pi (5 - x^2)^2 - \pi (5 - \sqrt{8x})^2 dx$

or $\int_0^4 2\pi (5 - y) (\sqrt{y} - \frac{1}{8} y^2) dy$

⑥ $m_y = \int_0^2 x (\sqrt{8x - x^2}) dx$

$m_x = \int_0^2 \frac{1}{2} (\sqrt{8x + x^2}) (\sqrt{8x - x^2}) dx$

② $\int_0^2 \sqrt{1 + \left(\frac{1}{2\sqrt{8x}} 8\right)^2} dx$

③ $\int_0^{2\pi} \frac{1}{2} (3(1 - \cos(\theta)))^2 d\theta$

④ ① $\begin{matrix} x^2 & \sin(3x) \\ 2x & -\frac{1}{3} \cos(3x) \\ 2 & -\frac{1}{9} \sin(3x) \\ & + \frac{1}{27} \cos(3x) \end{matrix}$ $-\frac{1}{3} x^2 \cos(3x) + \frac{-1}{9} 2x \sin(3x) + \frac{2}{27} \cos(3x) + C$

⑤ $\frac{A}{x-3} + \frac{B}{(x-3)^2}$

$A(x-3) + B = 2x$
 $B = 6$
 $A = 2$

$2 \ln|x-3| + 6 \frac{(x-3)^{-1}}{-1} + C$

⑥ $\int \frac{\csc(\ln 3x)}{2x} dx = \frac{1}{2} \int \csc u du = -\frac{1}{2} \ln |\csc(\ln 3x) + \cot(\ln 3x)| + C$

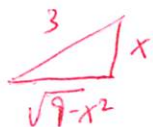
$u = \ln 3x$

$du = \frac{1}{x} dx$

$$\textcircled{D} \int \frac{x^2}{\sqrt{9-x^2}} dx = \int \frac{9 \sin^2 \theta}{3 \cos \theta} 3 \cos \theta d\theta$$

$$x = 3 \sin \theta$$

$$dx = 3 \cos \theta d\theta$$



$$\frac{9}{2} \int 1 - \cos(2\theta) d\theta = \frac{9}{2} \left(\theta - \frac{1}{2} \sin(2\theta) \right) + C$$

$$= \frac{9}{2} (\theta - \sin \theta \cos \theta) + C$$

$$\frac{9}{2} \left(\sin^{-1}\left(\frac{x}{3}\right) - \frac{x}{3} \frac{\sqrt{9-x^2}}{3} \right) + C$$

$$\textcircled{5} \textcircled{A} \sum \frac{\sqrt{k-3}}{k^2} \quad \text{LCT} \quad \lim_{k \rightarrow \infty} \frac{\frac{\sqrt{k-3}}{k^2}}{\frac{1}{k^{1.5}}} = 1 \quad \text{conv.}$$

$$\textcircled{B} \sum \frac{(2k)!}{e^k} \quad \text{div by D.v. Test} \quad \lim_{k \rightarrow \infty} \frac{\frac{(2k+2)!}{e^{k+1}}}{\frac{(2k)!}{e^k}} = \infty \quad \text{Ratio Test}$$

$$\textcircled{C} \sum \frac{\sin(3k)}{k!} \quad \left| \frac{\sin(3k)}{k!} \right| < \frac{1}{k!} \quad \text{conv. abs. b/c } \sum \frac{1}{k!} \text{ conv. (to } e^1 \text{) by Ratio Test etc.}$$

$$\textcircled{D} \sum_{k=1}^{\infty} \frac{2}{k} - \frac{2}{k+1} = \left(\frac{2}{2} - \frac{2}{3} \right) + \left(\frac{2}{3} - \frac{2}{4} \right) + \dots \quad \text{conv. to } 1$$

$$\textcircled{6} \sum \frac{2}{3^k k} (x-4)^k \quad \lim_{k \rightarrow \infty} \frac{\frac{2}{3^{k+1} (k+1)} (x-4)^{k+1}}{\frac{2}{3^k k} (x-4)^k} = \frac{|x-4|}{3} < 1 \Rightarrow 1 < x < 7$$

check endpoints $x=7$ $\sum \frac{2}{k}$ div. $R=3$

$x=1$ $\sum \frac{2(-1)^k}{k}$ conv. $[1, 7)$

$$\textcircled{7} \quad f(x) = \sqrt{x+3}$$

$$f(0) = \sqrt{3}$$

$$f'(x) = \frac{1}{2\sqrt{x+3}}$$

$$f'(0) = \frac{1}{2\sqrt{3}}$$

$$f''(x) = \frac{-1}{4} (x+3)^{-3/2}$$

$$f''(0) = \frac{-1}{(4) 3^{3/2}}$$

$$\sqrt{3} + \frac{1}{2\sqrt{3}} X + \frac{\frac{-1}{4} 3^{3/2}}{2} X^2$$

$\textcircled{8}$

$$u = \langle 1, 0, -2 \rangle$$

$$v = \langle 2, -3, 0 \rangle$$

$$\textcircled{A} \quad u - 3v = \langle -5, 9, -2 \rangle$$

$\textcircled{B}$

$$2 = u \cdot v = \|u\| \|v\| \cos \theta = \sqrt{5} \sqrt{13} \cos \theta$$

$$\cos \theta = \frac{2}{\sqrt{5} \sqrt{13}}$$

$\textcircled{C}$



$$\text{Area} = \|u \times v\| \cdot \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 0 & -2 \\ 2 & -3 & 0 \end{vmatrix} = \frac{-6\vec{i} - 4\vec{j} - 3\vec{k}}{\sqrt{61}}$$